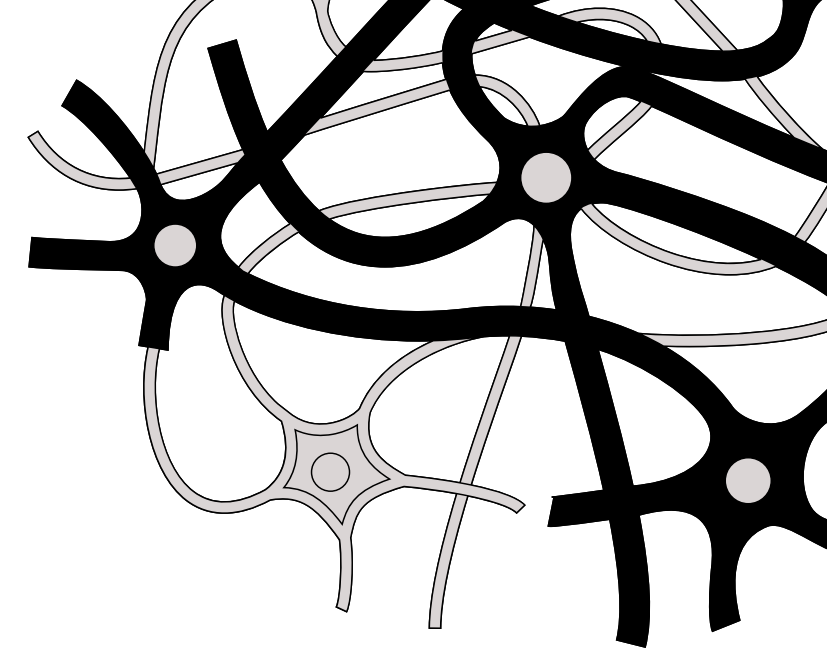
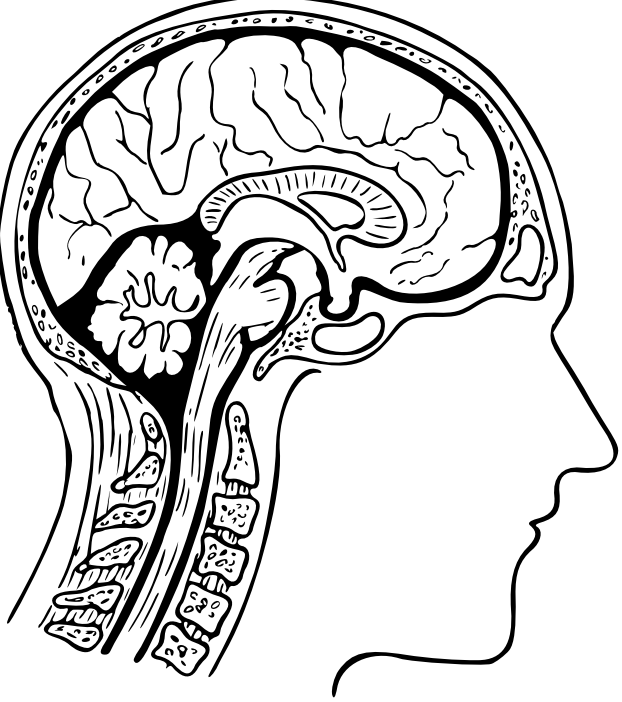
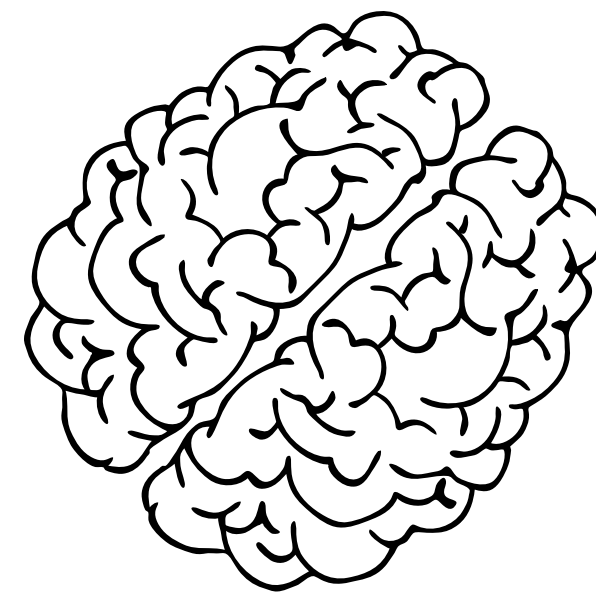
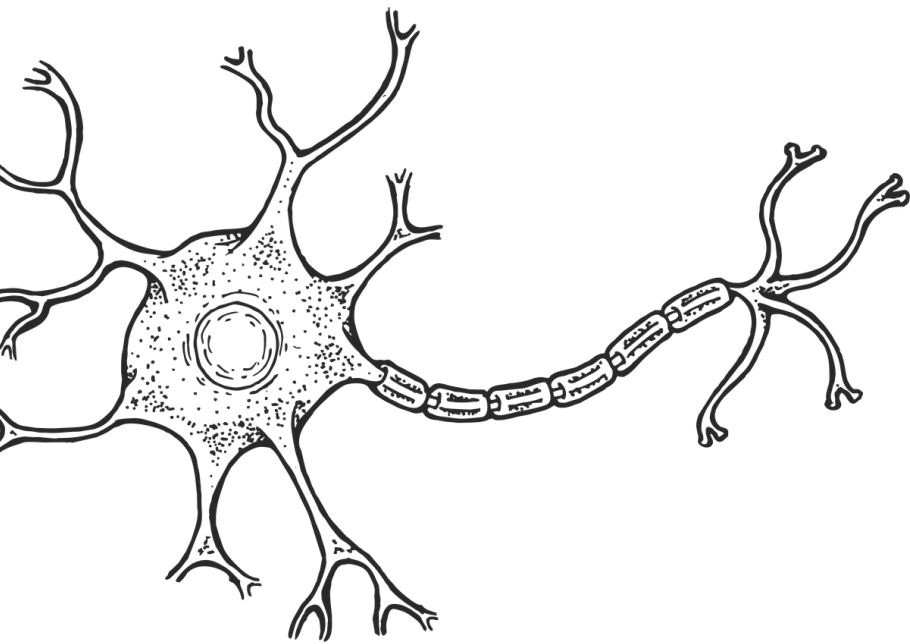
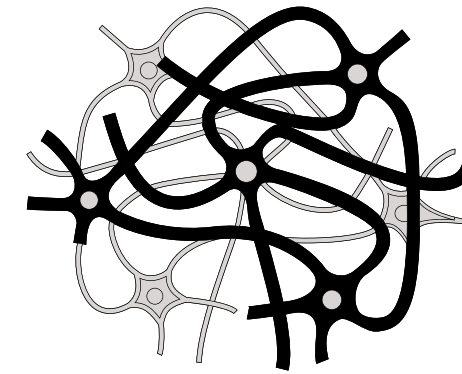
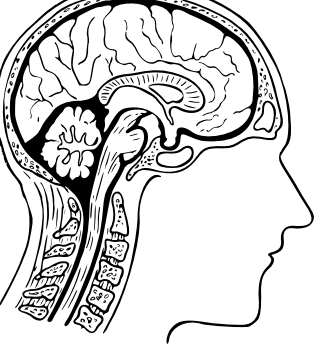




Session 9: ERP Fundamentals

presented by Lynn





ROADMAP

Cognitive

Analyze the cognitive processes involved in ERP responses.



Sensory

Explore higher-order neural computations and their implications.



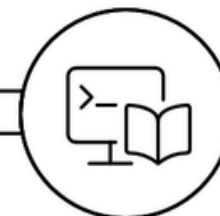
Theory

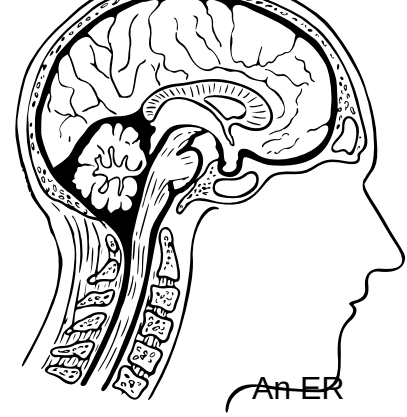
Focus on understanding the early visual and auditory processing stages.



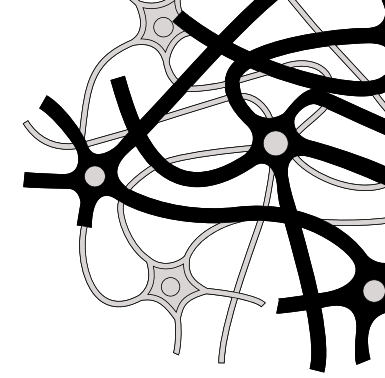
Code

Implement MNE Python for live coding and identification.





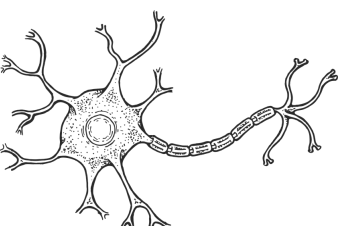
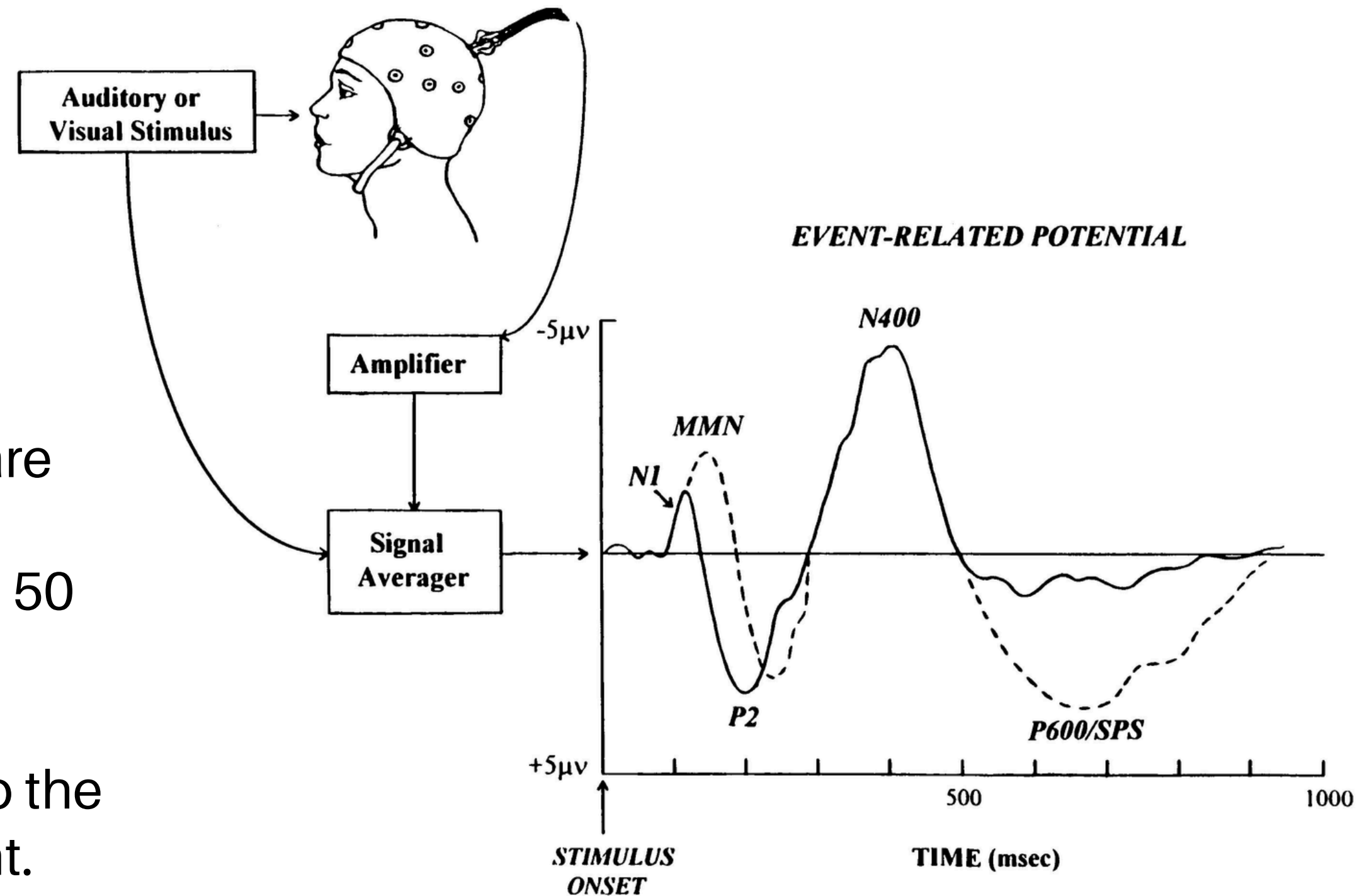
Theory: Defining ERP

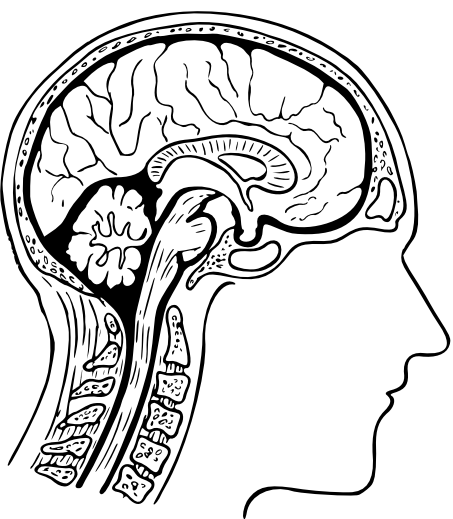


- Concept

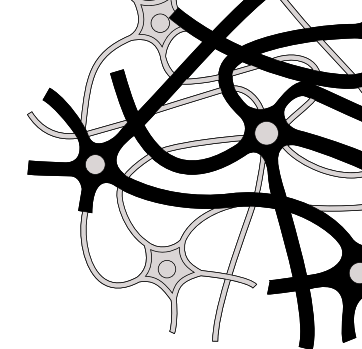
- An ERP is an averaged EEG waveform that is strictly time-locked to a specific sensory, motor, or cognitive event

- **The Problem:** Neural signals are tiny ($\approx 5 \mu\text{V}$). Spontaneous background EEG is massive ($\approx 50 \mu\text{V}$).
- **Solution:** Signal Averaging.
- **Signal:** The neural response to the stimulus is fixed and consistent.
- **Noise:** Background brain activity is random relative to the stimulus.





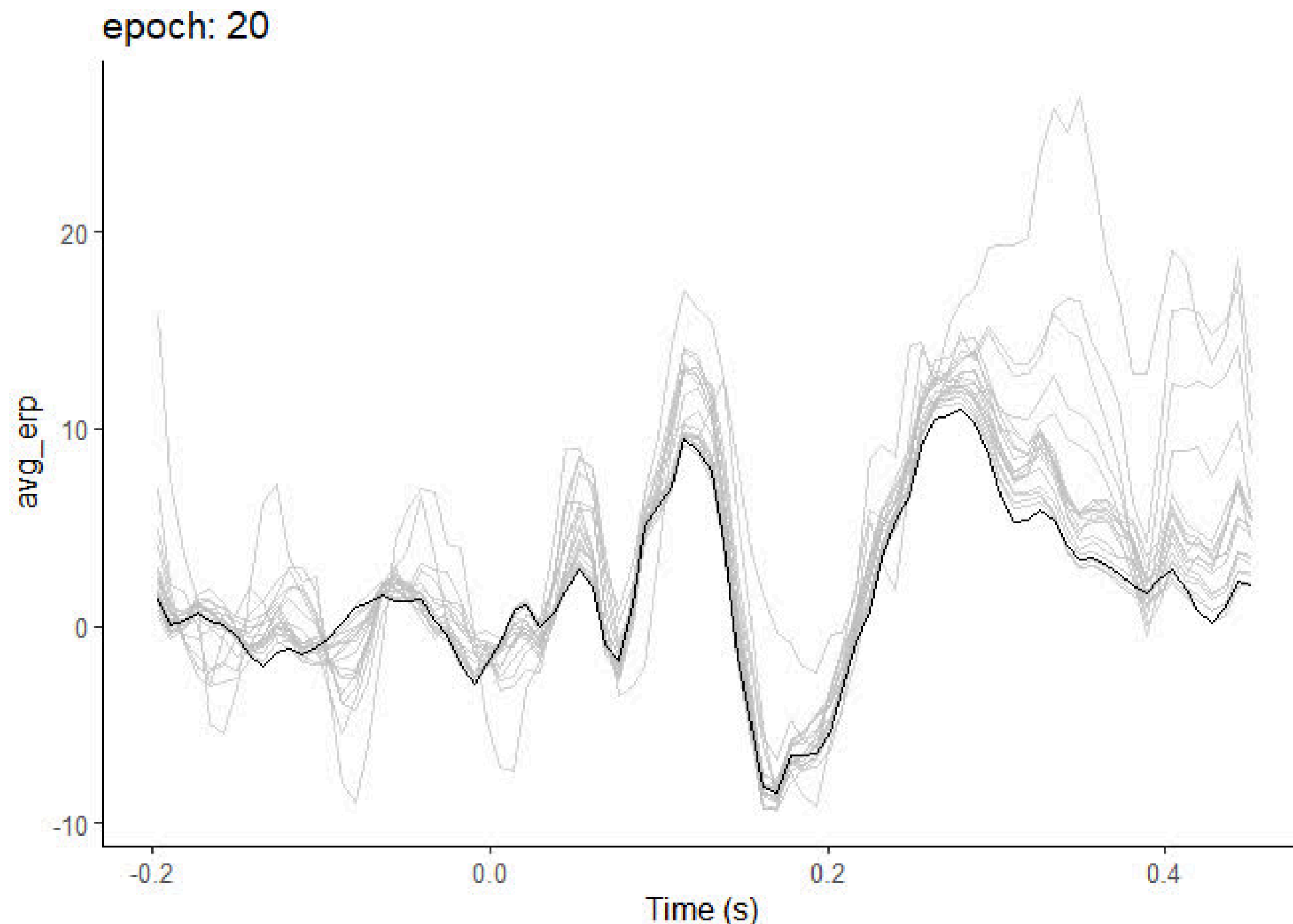
Signal-to-Noise Maths



To extract the signal, we rely on the maths of signal averaging. Random background noise destructively interferes and vanishes over many repetitions

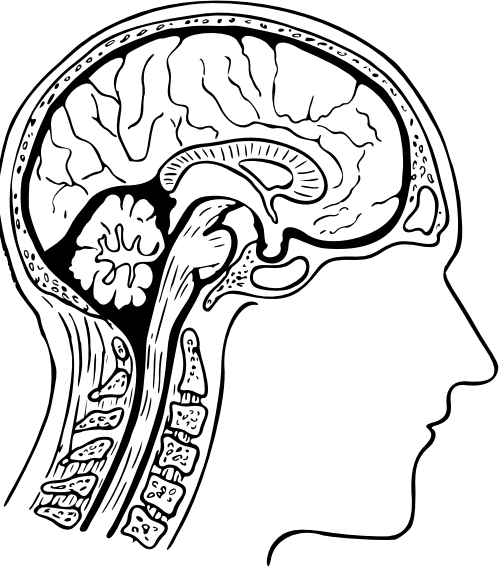
$$SNR \propto \sqrt{N}$$

- **The Math:** Signal-to-Noise Ratio improves in proportion to the square root of the number of trials (N).
- **The Rule:** Robust ERP designs require 20 to 100+ trials per condition to isolate the component.

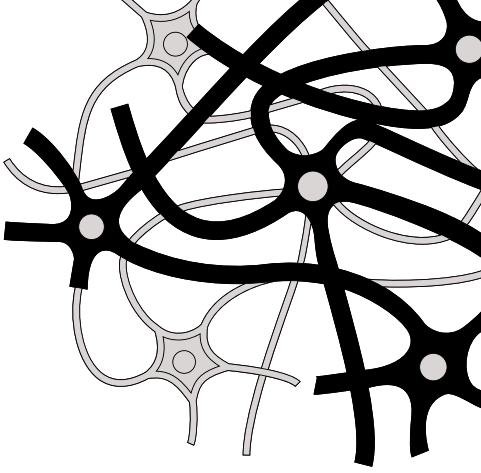


[Building up an ERP](#)





Reading the Waveform



ERP components are classified by their physical morphology across two key dimensions.

Polarity

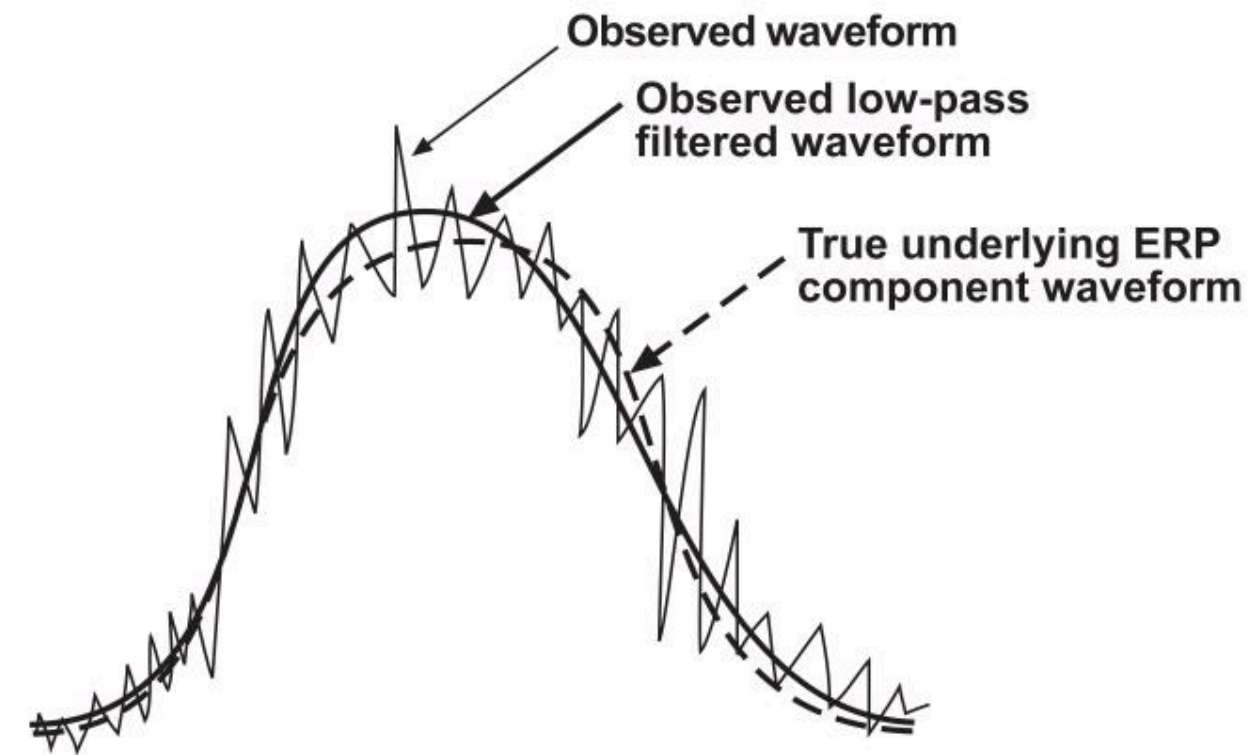
- +P = Positive-going voltage.
- -N = Negative-going voltage.

Note: Older literature often plots negative voltages pointing UP. Always check the Y-axis.

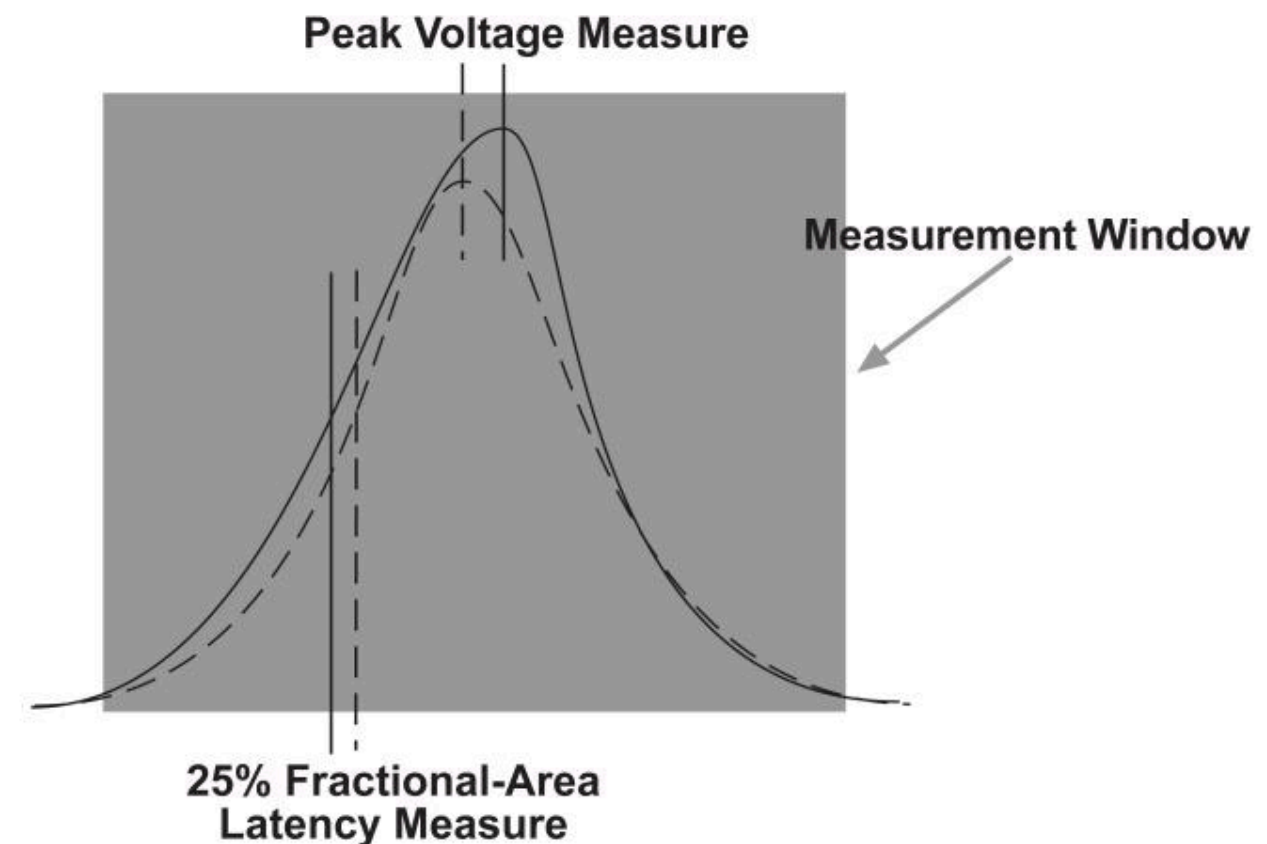
Latency

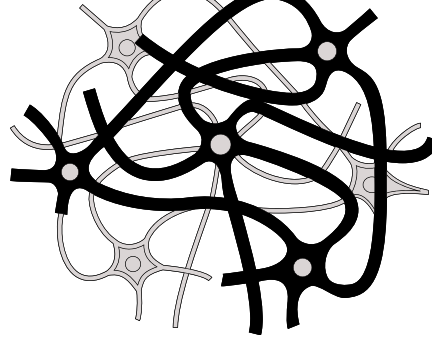
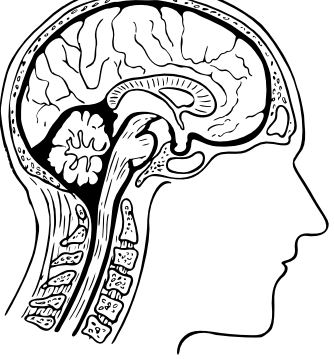
Denoted by sequence (e.g., P1, N1) or approximate timing in milliseconds (e.g., P300, N400).

A



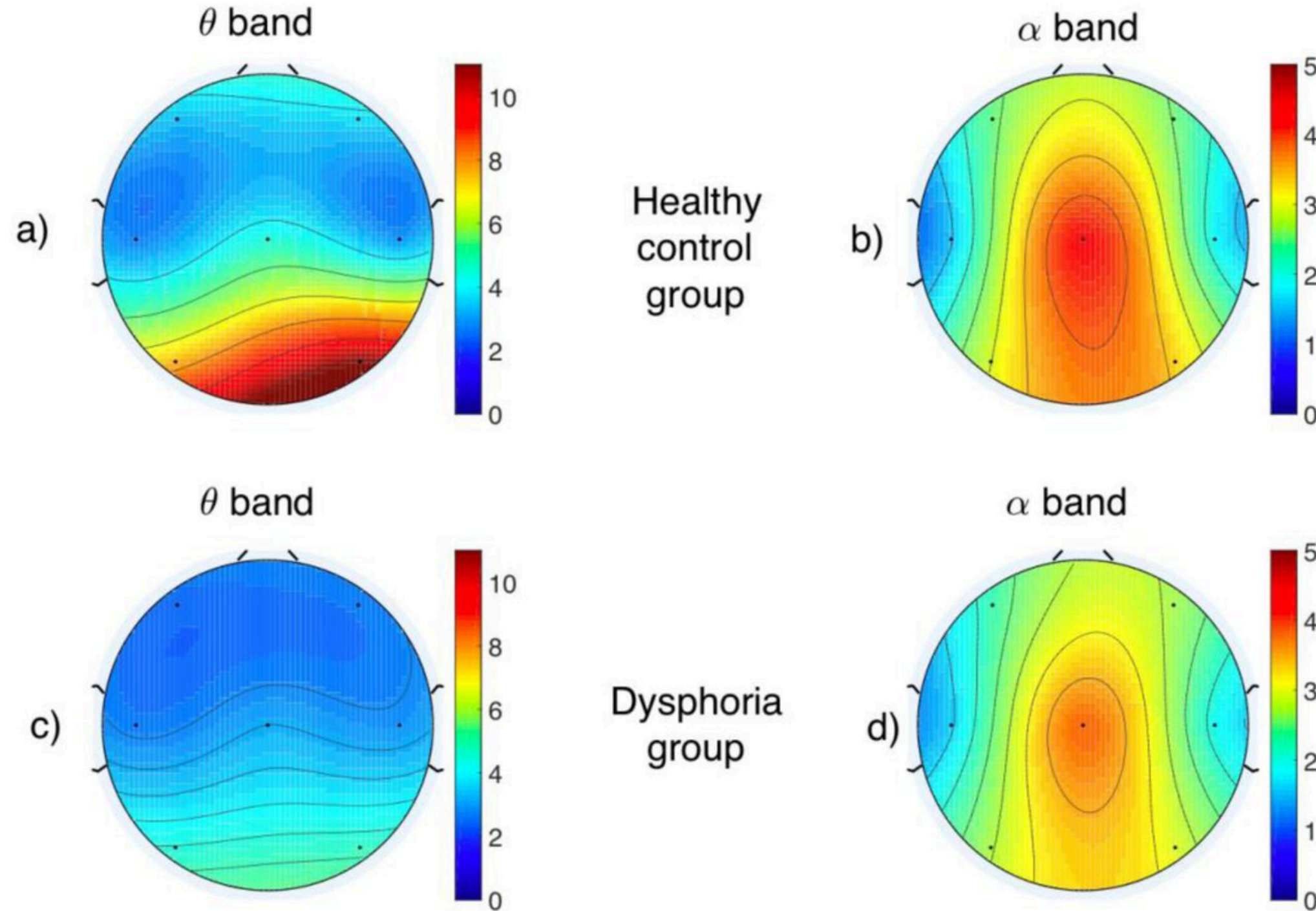
B





Sensory: Obligatory Processing

First 200ms



P1 (80-120 ms)

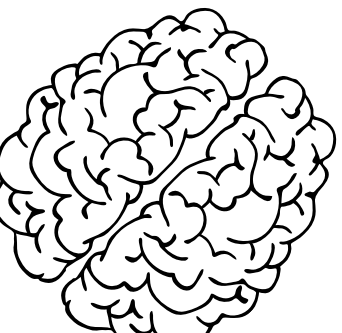
Scalp Maximum: Occipital lobe.

Function: Early visual sensory processing. Highly sensitive to spatial attention (larger amplitude for attended locations).

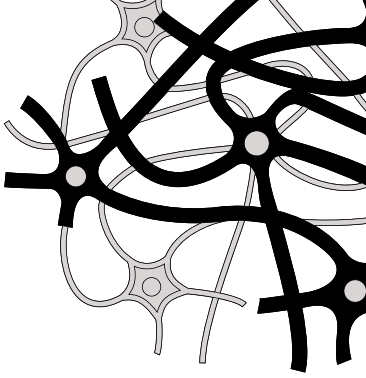
N1 (90-200 ms)

Scalp Maximum: Occipital and Frontocentral.

Function: Sensory discrimination and orienting of attention.



Feature Extraction



P2 (150–250 ms)

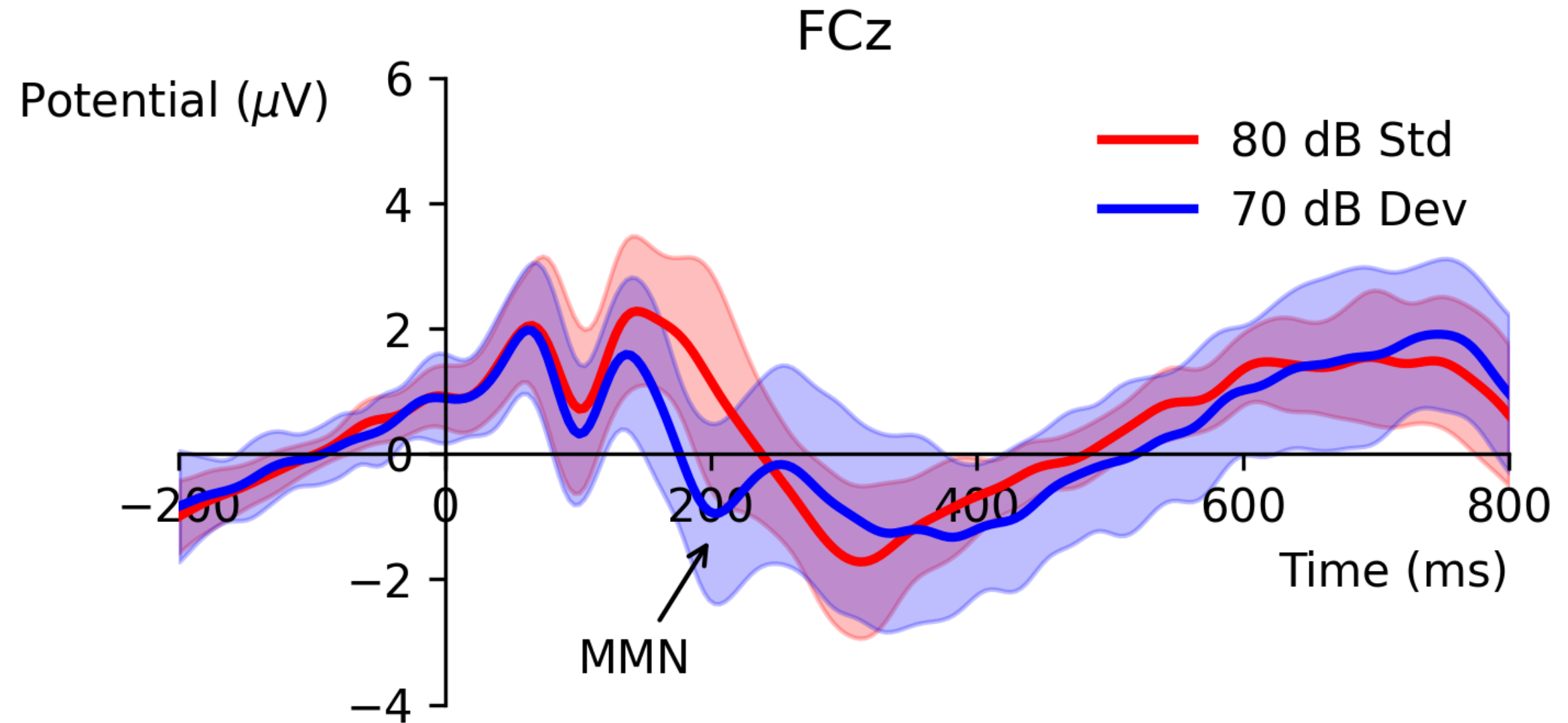
Scalp Maximum: Frontocentral.

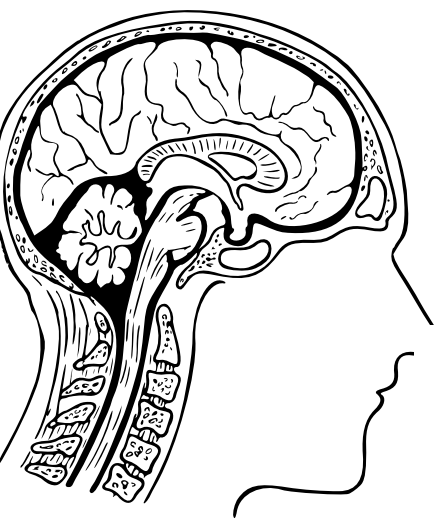
Function: Higher-order stimulus categorization and early feature detection mechanisms.

MMN (Mismatch Negativity)

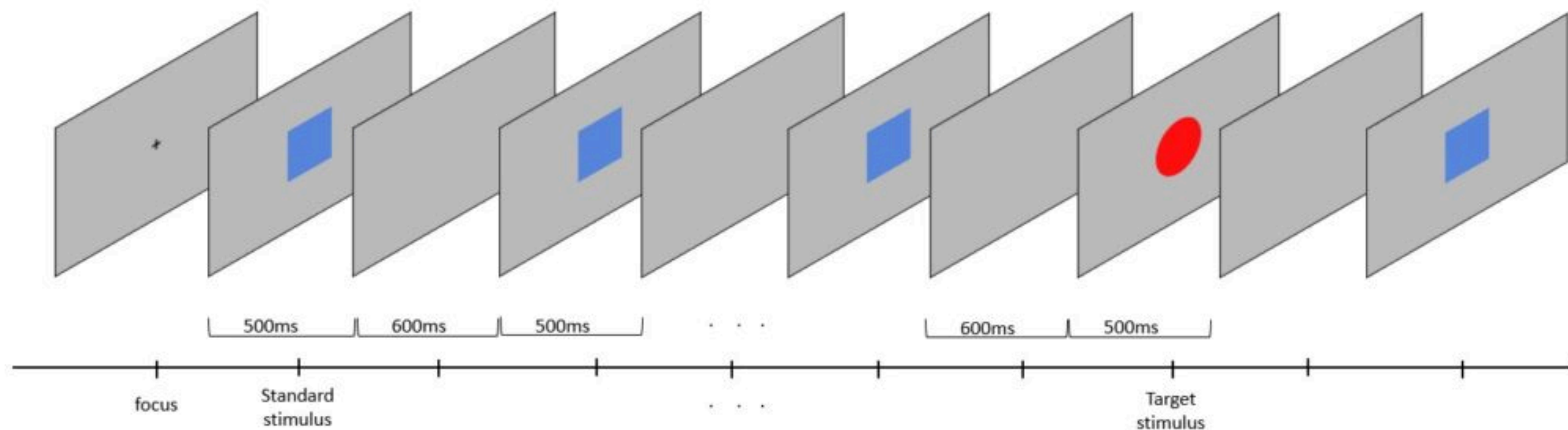
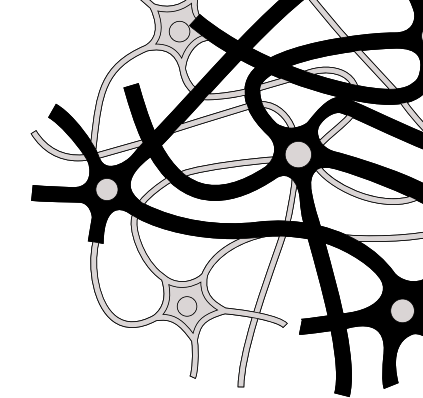
Function: A pre-attentive marker of auditory change detection.

Triggers when a pattern is broken, even if the subject is actively ignoring the stimulus sequence.





Cognitive: Context Updating



Cognitive Control

Higher order processes occurring after initial sensory extraction.

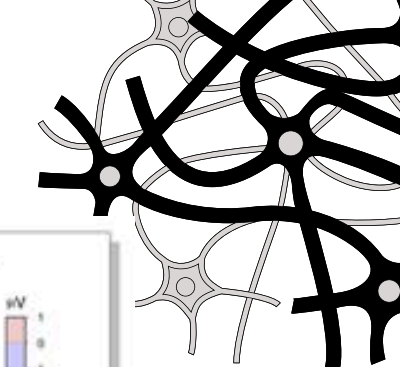
N2 (200–350 ms)

- **Location:** Frontocentral.
- **Function:** Conflict detection and motor inhibition (e.g., stopping a planned action).

P3 / P300 (300–600 ms)

- **Location:** Parietal (Electrode Pz).
- **Function:** Context updating, target detection, and working memory allocation.
- **Oddball Paradigm:** Elicited by a rare target hidden among frequent standard stimuli.

Semantic Processing



N400 (300–500 ms)

Scalp Maximum: Centroparietal.

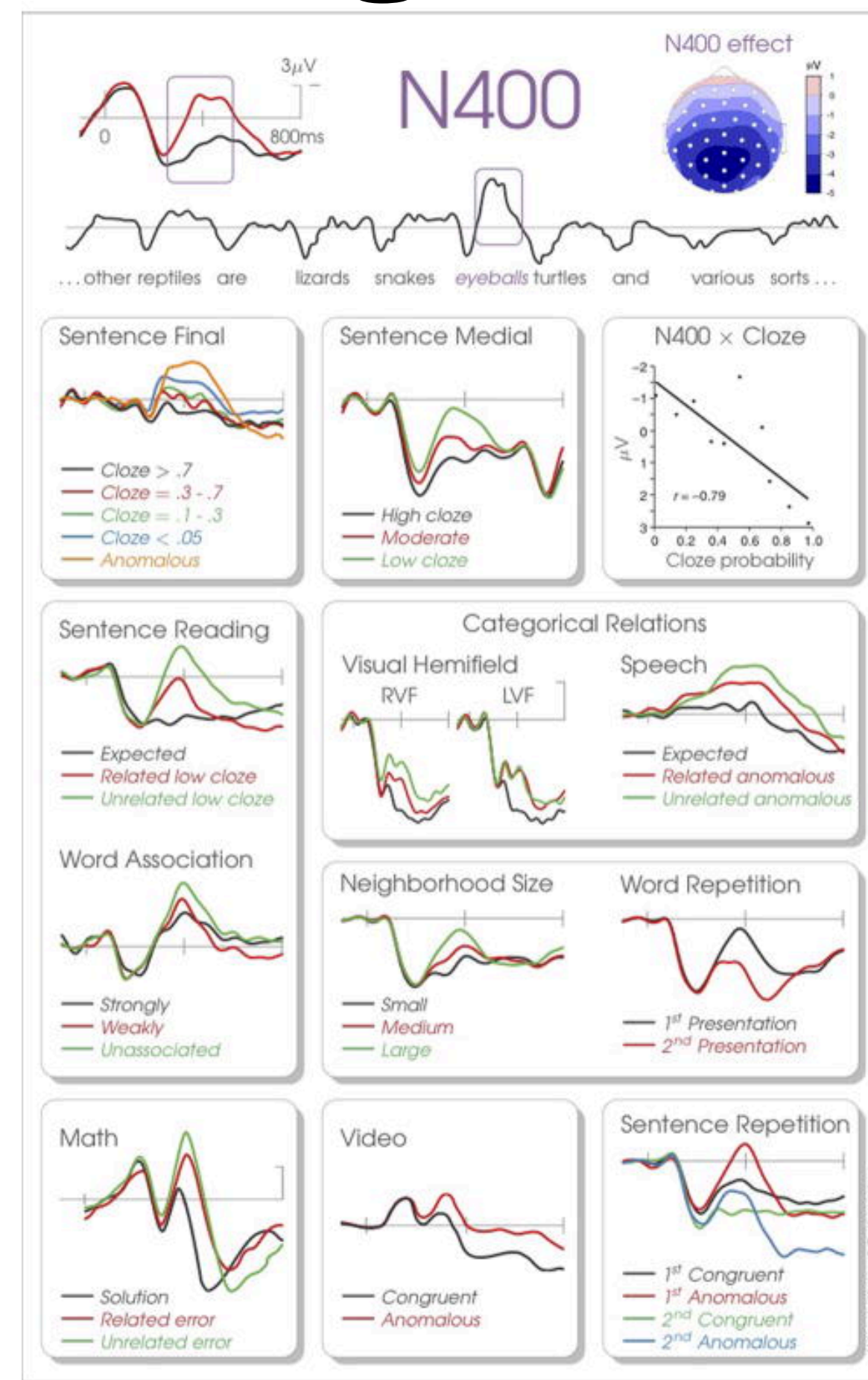
Function: Semantic processing and language expectation violation.

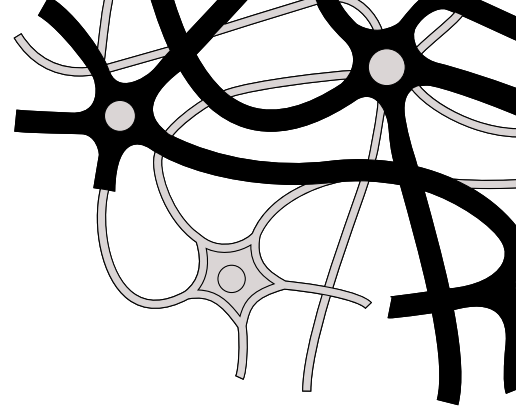
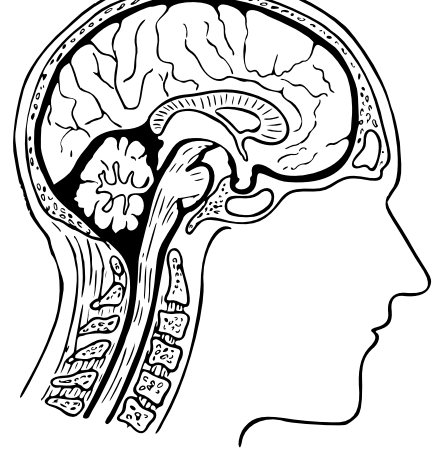
Kutas & Hillyard (1980)

The amplitude of the N400 scales directly with semantic incongruity.

The brain generates a massive negative voltage when struggling to integrate a word that completely violates contextual expectations.

"He spread his toast with socks."





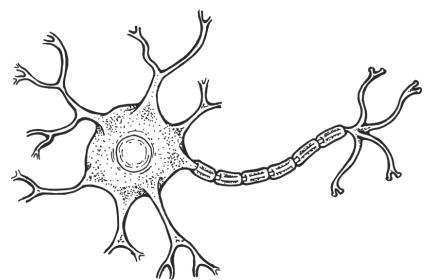
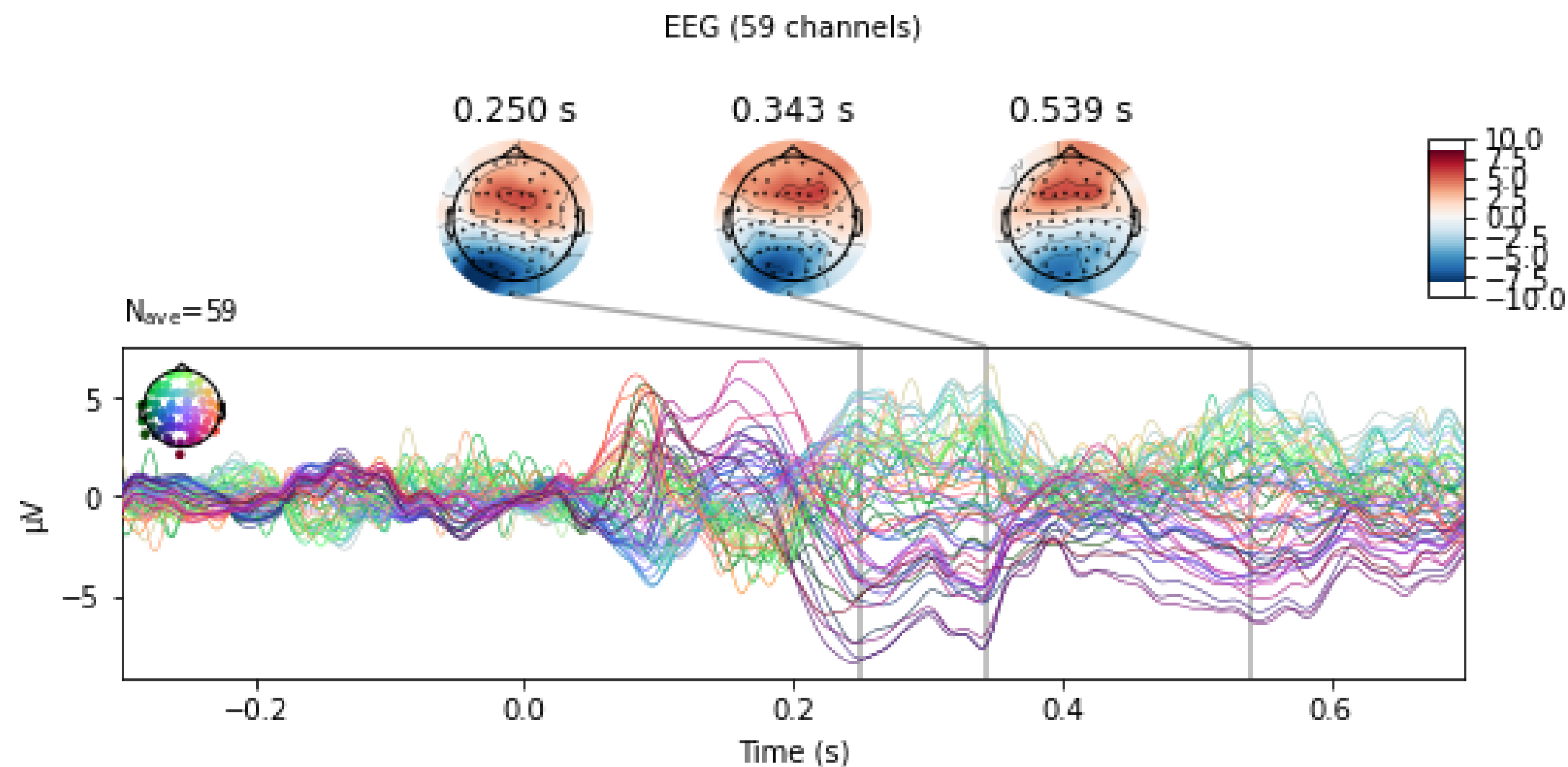
Code: MNE Implementation

The Task:

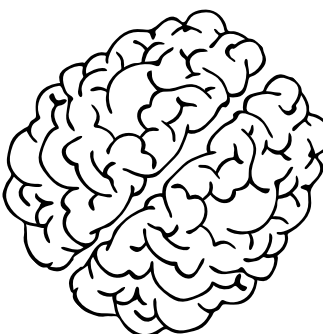
- Define epochs with proper baseline correction.
- Calculate the Evoked average using `.average()`.
- Plot the joint topographies.

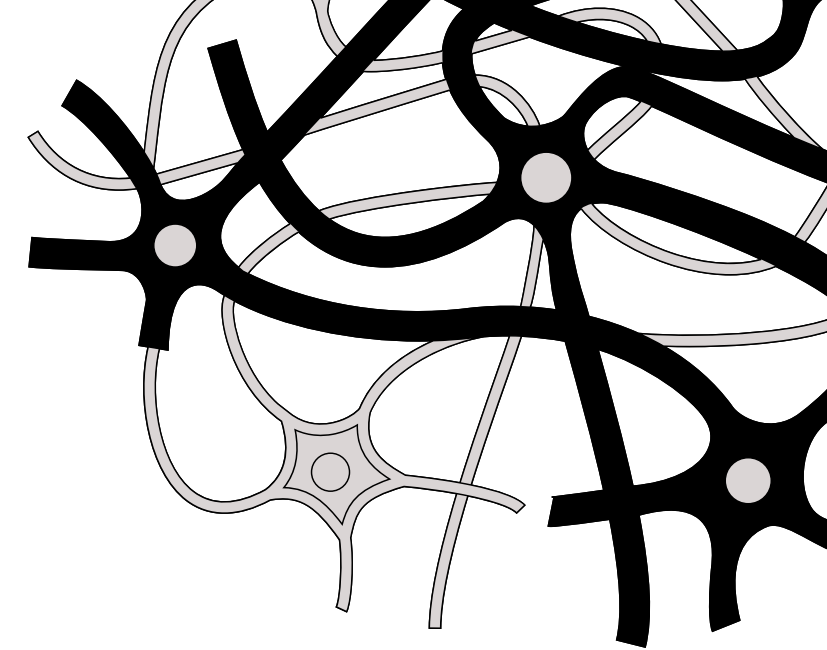
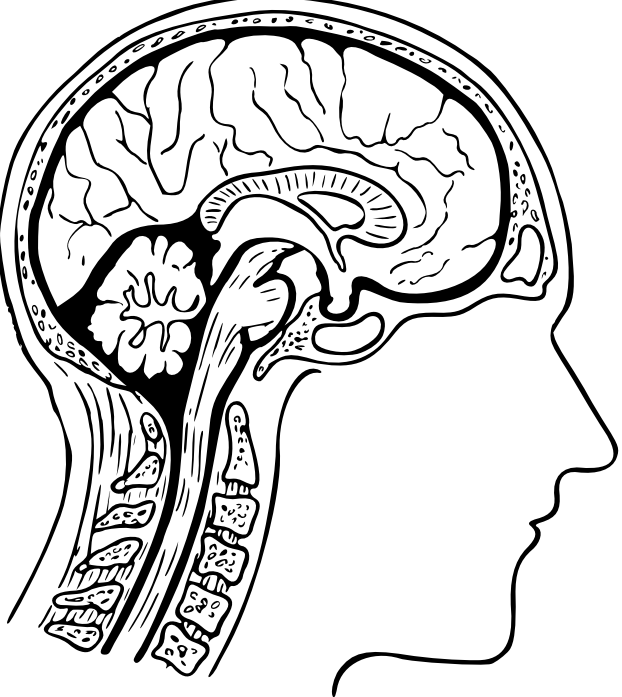
The Goal:

Given the generated figure on the right, independently identify the peaks, measure their latency and amplitude, and name the likely components based on today's theory.



[Colab](#)





**Thank
You**

